

Lecture #6

Content to be covered

- ❑ Michelson interferometer
 - ❖ Schematic and Working principle
 - ❖ Circular fringes and their radii
 - ❖ Role of Compensating Plate
- ❑ Applications of Michelson interferometer
 - ❖ Wavelength determination
 - ❖ Refractive index determination
 - ❖ Thickness of a transparent plate
 - ❖ Distinction between two neighbouring spectral lines

Resources to be consulted

- ❑ OPTICS by Ajay Ghatak

Michelson interferometer

Interferometer produces interference fringes by splitting a beam of monochromatic light so that one beam strikes a fixed mirror and the other a movable mirror. When the reflected beams are brought back together, an interference pattern results.

Michelson designed an interferometer to obtain interference fringes of various shapes which have a number of applications in optics. He was awarded the 1907 Nobel Prize in Physics for his “optical precision instruments and the spectroscopic and metrological investigations”.

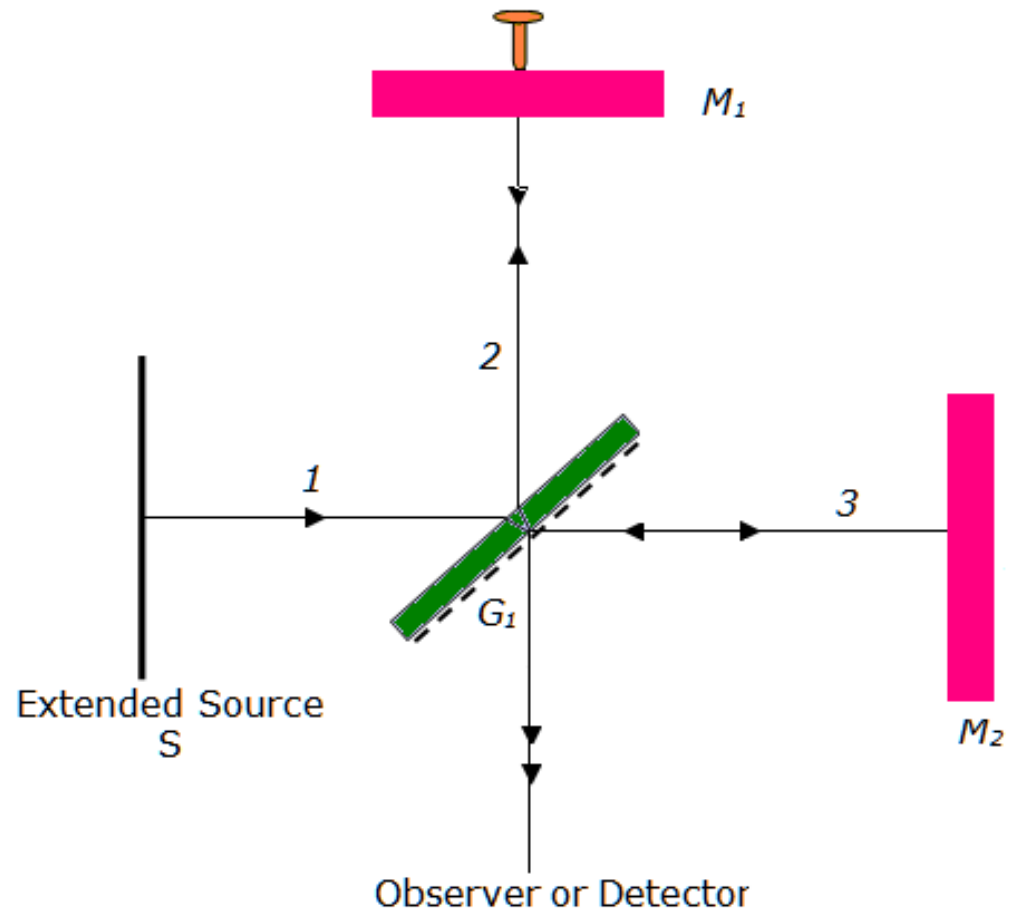
It can be used

1. To determine the wavelength of light.
2. To determine the refractive index and thickness of a glass plate.
3. To determine the difference in wavelength between two neighboring spectral lines.
4. Standardization of the Meter: 1 meter = 1553164.13 red cadmium wavelength (643.84 nm)

Michelson interferometer : Schematic

It has two plane mirrors and a glass plate G_1 . The plane mirrors are mounted vertically on two arms at right angles to each other. The mirror M_1 can be moved with the help of screws. The plate G_1 is inclined at 45° with M_1 and M_2 . The surface of G_1 towards M_2 is partially silvered (or glass plate).

1. monochromatic source
2. a detector
3. two mirrors
4. one semitransparent mirror (beam splitter)



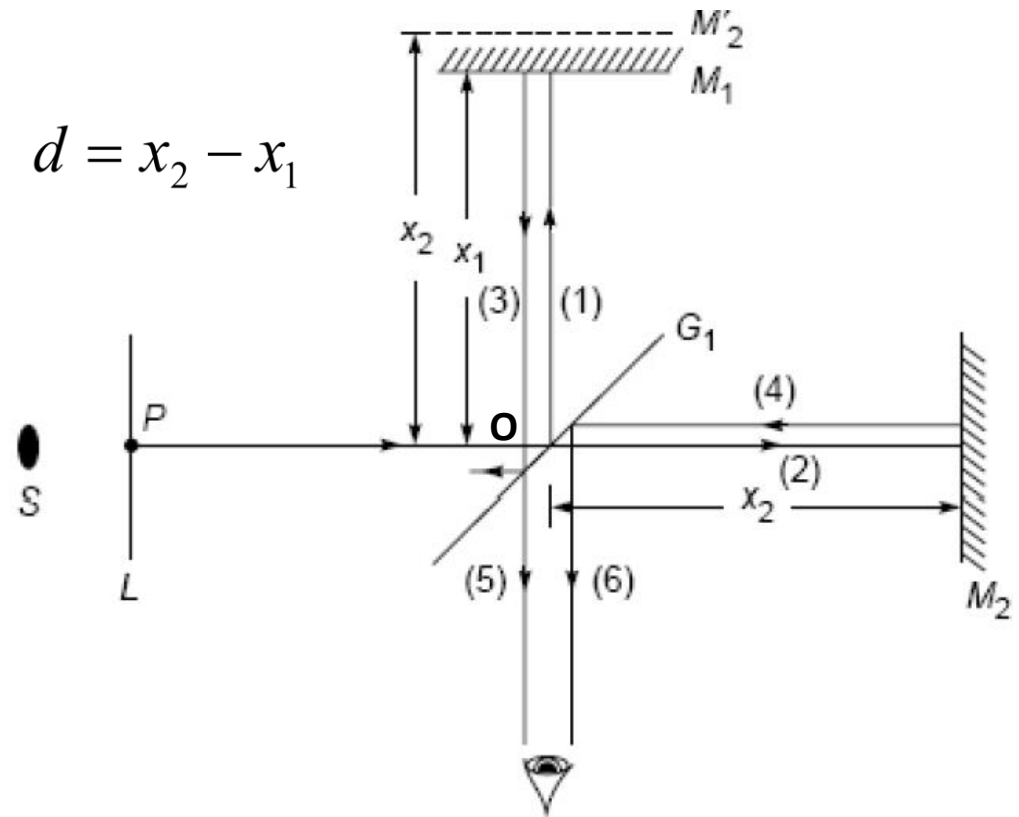
Michelson interferometer : Working Principle

❑ The form of fringes depends on the inclination of M_1 and M_2 .

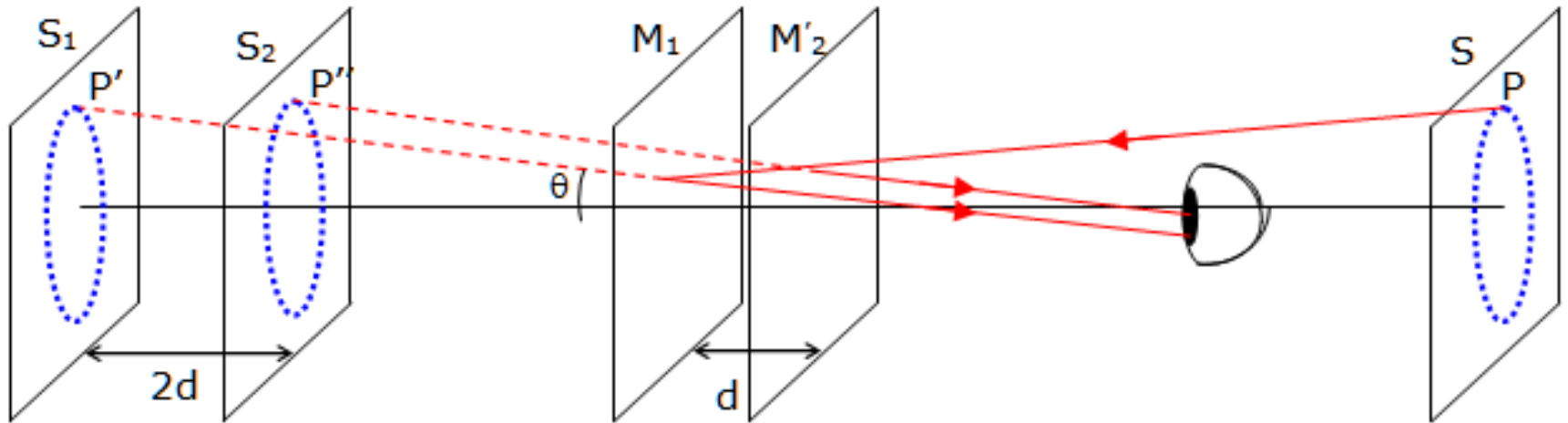
❑ Let M'_2 be the image of M_2 formed by reflection at semi-silvered surface of G_1 so that $OM'_2 = OM_2$

❑ The interference fringes may be regarded to be formed by light reflected from the surfaces of M_1 and M'_2 .

❑ Thus the arrangement is equivalent to air film enclosed between the reflecting surfaces M_1 and M'_2 .

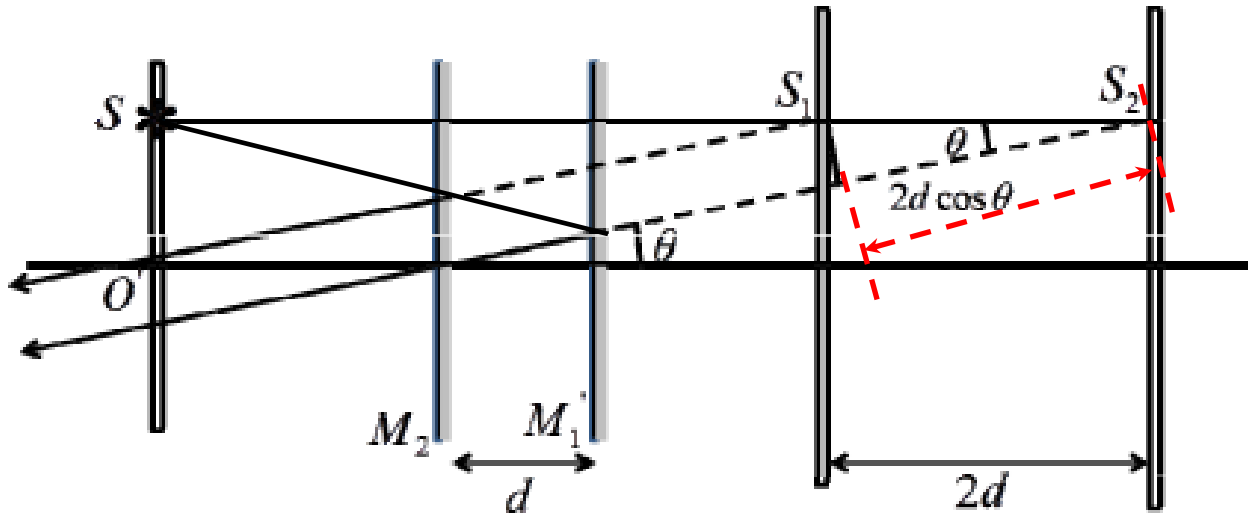


Equivalent optical set-up of Michelson Interferometer



- ❑ The real mirror M_2 is replaced by its virtual image M'_2 formed by reflection in G_1 .
- ❑ If M_1 and M_2 are exactly perpendicular to each other and G_1 is inclined at 45° to them, then M_1 and M'_2 are exactly parallel to each other.
- ❑ To the observer, the two rays will appear to be originating from the virtual images (S_1 & S_2). These virtual sources are coherent sources.
- ❑ The rays coming from these virtual images are parallel to each other because M_1 is parallel to M'_2 .

Formation of Fringes



$$\Delta = 2d \cos \theta$$

$\nearrow$ fix $\nearrow$ variable

If the beam splitter is just a simple glass plate, the beam reflected from mirror M_2 will undergo an abrupt phase change of π (when getting reflected by the beam splitter).

The positions of minimum and maximum intensity in the interference pattern is given by

Minima $2d \cos \theta = n\lambda$

Maxima $2d \cos \theta = (2n + 1)\frac{\lambda}{2}$

Circular Fringes

For dark rings:

$$2d \cos \theta = n\lambda$$

Consider $\lambda = 5 \times 10^{-5}$ cm & $d = 0.025$ cm then the angles θ where the dark fringes occur are given by:

$$\theta = \cos^{-1} \frac{n\lambda}{2d}$$

$$\theta = \cos^{-1}(0.001n)$$

$$\theta = 0, 2.56, 3.62, 4.44, 5.13, 5.73, 6.28, \dots$$

$$n = 1000, 999, 998, 997, \dots$$

If, $d = 0.0025$ cm, then

$$\theta = 0, 8.11, 11.48, 14.07, 16.26, \dots$$

$$n = 100, 99, 98, 97, \dots$$

So, **as d decreases the fringes will appear to collapse at the centre.**

The highest order fringe appears at the centre (i.e. corresponding to $\theta = 0^\circ$)

If, d slightly decreases (let from 0.025 cm to 0.024999 cm) then

$$\theta = 2.51, 3.59, 4.41, \dots$$

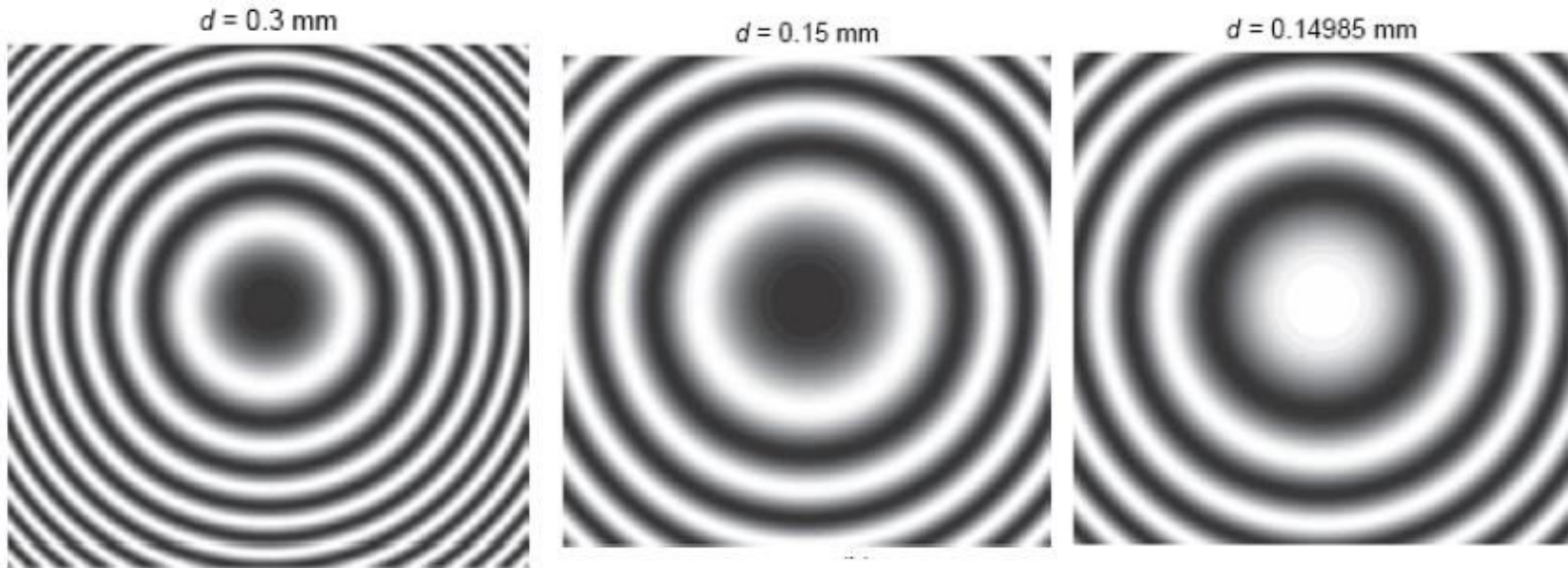
$$n = 999, 998, 997, \dots$$

Here, **$n = 1000$ disappears**

This indicates that as d decreases, fringes become less closely spaced.

Circular Fringes

When M_2 is exactly perpendicular to M_1 , the film $M_1 M'_2$ is of uniform thickness and we obtain circular fringes **localized at infinity**.



- ✓ also known as Haidinger fringes
- ✓ fringes localized at infinity
- ✓ fringes of equal inclination
- ✓ They can be observed by introducing a converging lens between the film and observation plane with focus of the lens lying in observation plane.

Radius of Circular Fringe

A central dark fringe for which $\theta = 0$ can be represented by
At a fixed value of d , successive dark rings will be given by

$$2d \cos \theta_1 = (n-1)\lambda$$

$$2d \cos \theta_2 = (n-2)\lambda$$

□

$$2d \cos \theta_p = (n-p)\lambda \quad \dots\dots\dots(2)$$

The angular position of p^{th} ring is determined by $2d(1 - \cos \theta_p) = p\lambda$

Here, p be the number index of a ring starting from the center having angular Inclination θ_p and radius r_p . Then

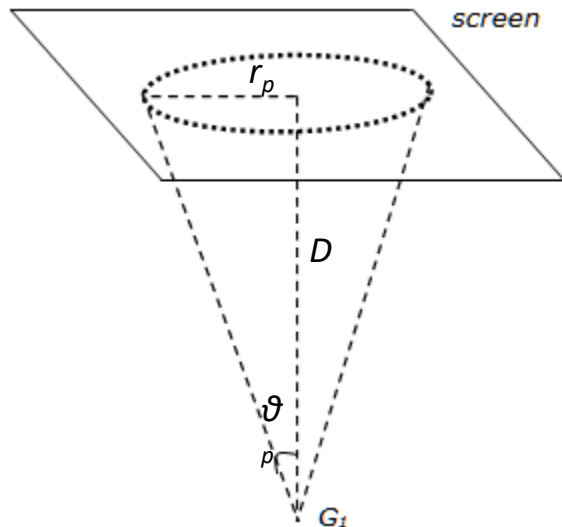
$$\theta_p^2 = \frac{p\lambda}{d}$$

$$\text{or } \theta_p \propto \sqrt{p}$$

If D is the separation between the glass plate G_1 and the Screen on which fringes are obtained, then $\theta_p = r_p/D$.

Hence,

$$r_p = D \sqrt{\frac{p\lambda}{d}}$$

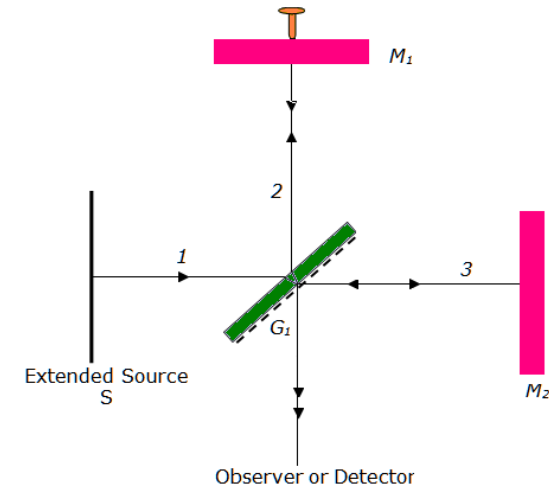


$$2d = n\lambda \quad \dots\dots\dots(1)$$

If $d = 1\text{cm}$ & $\lambda = 500\text{ nm}$,
 n_0 will be 40,000.

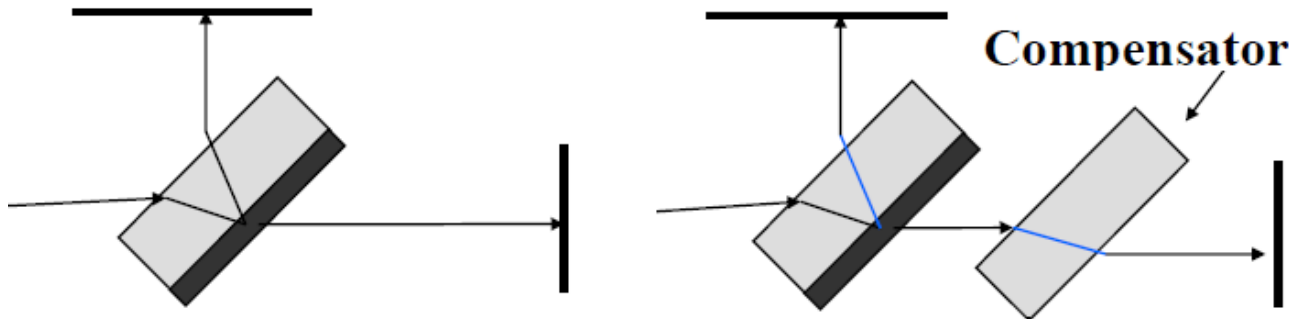
Role of Compensating Plate

Beam #2 traverse the glass plate G_1 thrice whereas beam #3 traverse it only once. Hence, the earlier beam travels an additional $2\mu t$ (μ : refractive index of glass plate G_1 for the wavelength of light used and t is the physical distance travelled by the ray through the plate). This extra path difference between the two interfering beams is a fixed quantity. This path difference is of great concern for white light due to dispersion.



❑ To compensate this, a compensating glass plate G_2 made of same material and dimension as G_1 is placed in the path of beam #3. This results in an extra optical path $2\mu t$ in the path of beam #3 also, which compensates for the additional path difference suffered by beam #2.

Any small inequalities in the optical paths can be removed by rotating the plate G_2 (varying the optical path through the thickness of the glass plate).



Let m fringes (bright or dark) collapse to the centre as the mirror M_2 moves by a distance d_0 .

initially for d $2d = n\lambda$ (1)

after changing d by d_0 $2(d - d_0) = (n \sim m)\lambda$ (2)

after subtraction

$$\lambda = \frac{2d_0}{m}$$

Example: If by moving M_2 through 2.9×10^{-2} cm, 1000 fringes collapses then

$$\lambda = \frac{2 \times 2.9 \times 10^{-2}}{1000} = 5.8 \times 10^{-5} \text{ cm}$$

This method was used by Michelson for the standardization of the meter. He had found that the red Cadmium line ($\lambda = 6438.4696 \text{ \AA}$) is one of the ideal monochromatic source.

1 meter = 1553164.125 red cadmium wavelengths, the accuracy is almost 1 part in 10^9 .

2. Determination of refractive index or thickness of a plate

Insert the given plate in the path of one of the interfering beams which introduces an extra optical path difference of $2(\mu-1)t$.

The fringes are therefore shifted. The movable mirror is now moved till the fringes are brought back to their initial position. If this displacement is x then

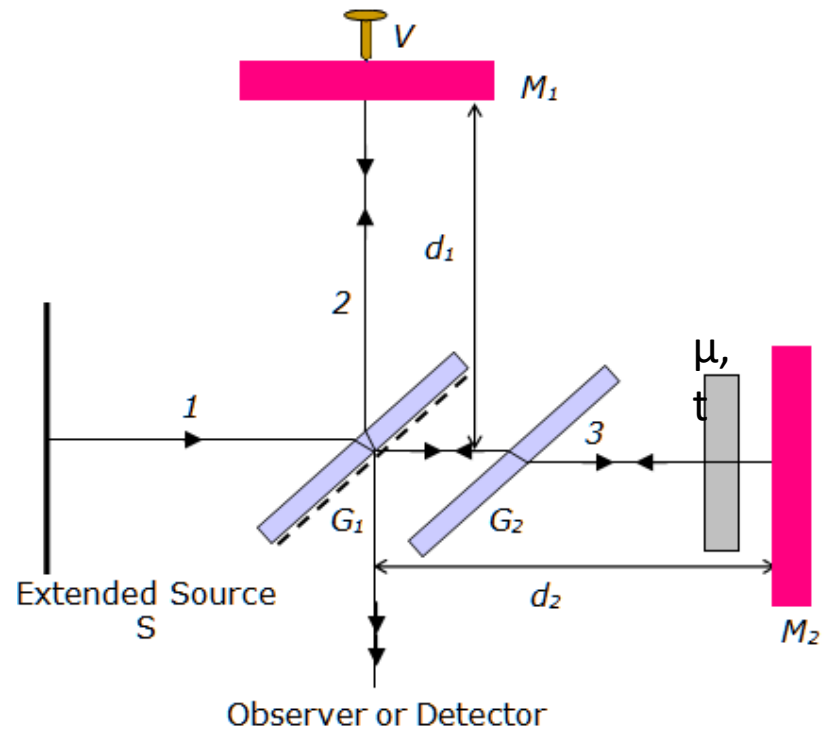
$$2x = 2(\mu-1)t$$

$$x = (\mu-1)t$$

$$2d = n\lambda$$

$$\& \ 2d + 2(\mu-1)t = m\lambda$$

$$2(\mu-1)t = (m-n)\lambda$$



3. Difference between two neighboring spectral lines

When $d = 0$, fringe pattern due to λ_1 and λ_2 would overlap. $n_1\lambda_1 = n_2\lambda_2$

If movable mirror is displaced by a distance d , then maxima corresponding to λ_1 will not coincide with that of λ_2 .

If $\lambda_1 > \lambda_2$ and $n_1 = n$ & $n_2 = n+1$ then, at $\theta = 0$, for dark fringes $n\lambda_1 = (n+1)\lambda_2$

and for bright fringes

$$(2n_1 + 1)\frac{\lambda_1}{2} = (2n_2 + 1)\frac{\lambda_2}{2}$$

$$\Rightarrow (2n + 1)\frac{\lambda_1}{2} = [2(n + 1) + 1]\frac{\lambda_2}{2}$$

These corresponds to the position of distinctness *i.e.* bright of λ_1 overlaps with bright of λ_2 which gives maximum visibility.

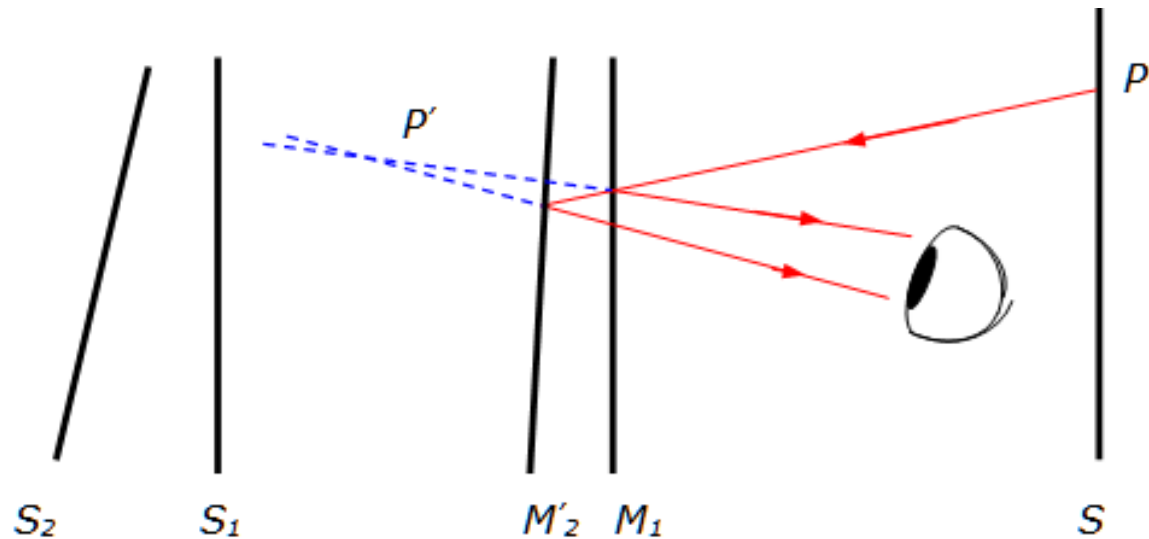
$$\Rightarrow n = \frac{\lambda_2}{\lambda_1 - \lambda_2} \Rightarrow 2d = n\lambda_1 = \frac{\lambda_1\lambda_2}{\lambda_1 - \lambda_2}$$

$$\lambda_1 - \lambda_2 = \frac{\lambda_1\lambda_2}{2d} = \frac{\lambda_{mean}^2}{2d}$$

$$\Delta\lambda = \lambda^2 / 2d$$

<u>Newton's Ring</u>	<u>Michelson's Interferometer</u>
$2\mu t \cos r = n\lambda$ $\uparrow$ variable $\nwarrow$ fix	$2d \cos \theta = n\lambda$ $\uparrow$ fix $\nwarrow$ variable
Circular fringes because of locus of equal thickness of film	Circular fringes because of equal inclination of light of film incidence
Newton's ring have minimum order at the centre	Michelson's interferometer have maximum order at the centre.

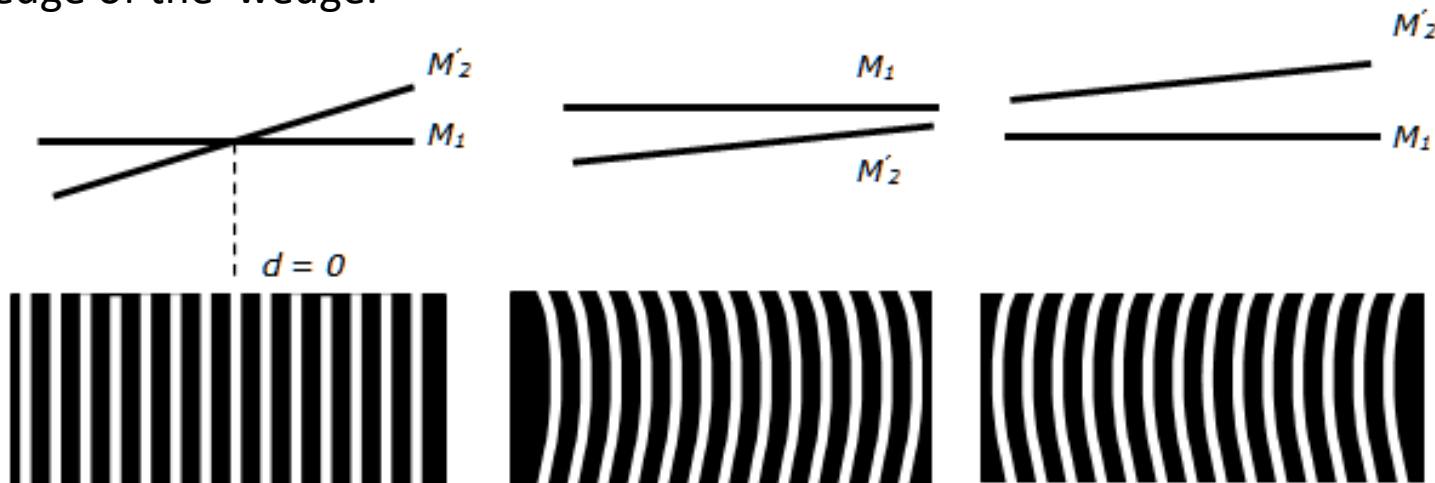
Localized Fringes



When the mirrors M_1 and M_2 are not perfectly perpendicular to each other and are inclined at an angle, then the two virtual sources (S_1 & S_2) formed in them are also inclined to each other. The space between them is like a wedge shaped film that results in formation of fringes of equal thickness. The two interfering rays are now no longer parallel but appear to diverge from a point P' behind the mirror M_1 . The eye must be focused on or near the rear side of M_1 to see these fringes clearly, and therefore these fringes are referred to as localized fringes.

Localized Fringes

- ❑ For **small angles of inclination between the two mirrors and for small path difference d** , these fringes are practically straight because locus of all point having same thickness of the wedge is a straight line parallel to the edge of the film.
- ❑ For larger values of d , the variation of angle of incidence from extended source also affects the path difference, & hence the fringes are slightly curved.
- ❑ When the path difference between the centers of the mirrors is zero, the M_1 and M'_2 intersect and the fringes are exactly straight.
- ❑ When M_2 is not perpendicular to M_1 , the air film between the M_1 and M'_2 is wedge shaped. As d increases, the fringes start to become curved and are convex towards the thin edge of the wedge.

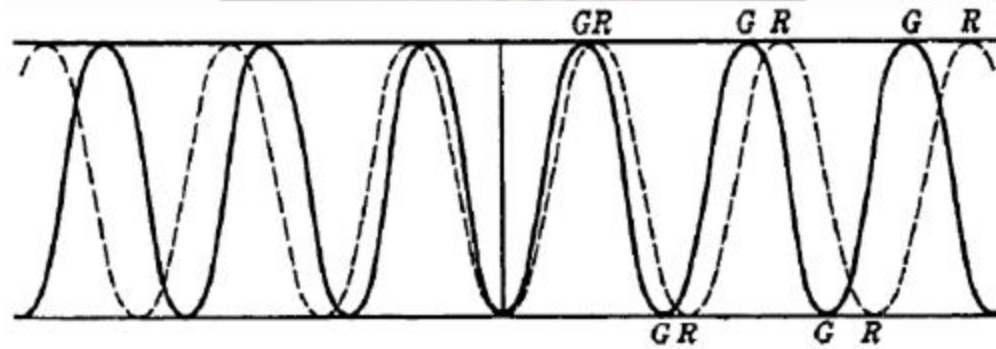
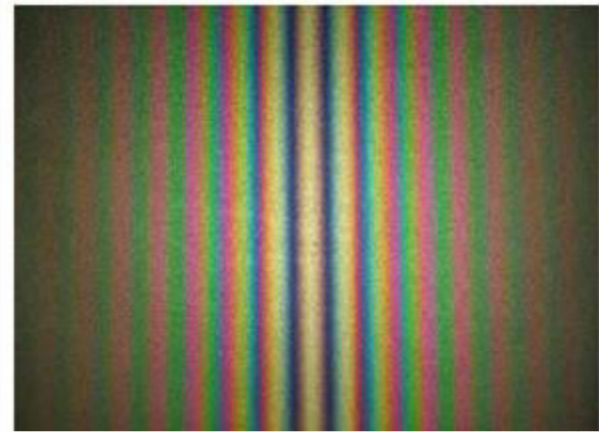


White-light Fringes

❑ If in the last case, monochromatic light is replaced by white light **and if thickness of the film is small (distance between M_1 and M_2')**, a few curved and colored localized fringes are obtained.

❑ The fringe of zero thickness will be perfect straight and perfect achromatic.

❑ For large thickness of the air film, uniform illumination is obtained.



Summary

- ❑ Michelson Interferometer is a very precise instrument based on the *division of amplitude* class of Interference.
- ❑ Light from an extended source is split along two arms of the interferometer using two highly reflecting mirrors and a 50% polished beam splitter, and is recombined to form interference pattern.
- ❑ The interference patterns so produced is very sensitive to micron-level optical path difference between two arms of the interferometer and hence this equipment can be used as a precise measuring tool.
- ❑ Different types of interference fringes (fringes of equal inclination, fringes of equal width *etc.*) can be obtained under different settings of the interferometer, and can be used for specific measuring applications.